Nicholas B. Pratt

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Education

University of California Merced, Merced, CA — August 2026 - Current
Quantitative and Systems Biology PhD
School of Natural Sciences

University at Buffalo, Buffalo, NY — May 2026
Bioinformatics & Computational Biology B.S.
School of Engineering and Applied Sciences

Research Experience

Doctoral Researcher — 08/26 – Present

Principal Investigator (s): Zhong Wang, Tomas Rube
Joint Genome Institute, Berkeley National Lab; Department of Applied Mathematics, University of California Merced

- Explore strategies for de novo generation of eukaryotic chromosomal segments using conditioned sequence generation and fragment consensus assembly, with maize chromosome arm 9S as a test case for eukaryote-scale generation using PlantGFM.
- Develop homology-aware sequence models for bacterial biosynthetic gene cluster annotation and terpene synthase product prediction using GenomeOcean and ESM-2 embeddings.
- Apply the biophysics-based model ProBound to BEND3 ChIP-seq data to infer DNA-binding motifs and characterize sequence preferences of its BD3 and BD4 DNA-binding domains.

Visiting Researcher — 06/26 – Present

Principal Investigator (s): Aziz Khan
Division of Biological and Life Sciences, Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI)

- Develop AskJASPAR, an AI-native interface using the Model Context Protocol (MCP) to integrate LLM-based agents with the JASPAR transcription factor binding motif database, enabling programmatic retrieval and analysis of regulatory sequence information.

Undergraduate Research Assistant — 12/25 – 06/26 & 01/24 – 12/24

Principal Investigator (s): Jaroslaw Zola, Zhong Wang,
SCoRe (Scalable Computing Research) Group
Department of Computer Science, University at Buffalo; *Joint Genome Institute, Berkeley National Lab*
Responsibilities:

- Design and execute area-under-the-learning-curve experiments to quantify the learnability of genomic sequence data using information-theoretic metrics and comparisons of real, shuffled, and random sequences.
- Retrain Oxford Nanopore's Remora modified-base detection model to identify oxidative DNA damage, specifically 8-oxoguanine, from nanopore sequencing data.
- Perform DNA extraction, library preparation, Oxford Nanopore sequencing, and basecalling using experimental and bioinformatics workflows.

Laboratory Technician 08/25 – 01/26
Principal Investigator: Charlotte Lindqvist
Department of Biological Sciences, University at Buffalo

- Paid research position working in close collaboration with the Albert Lab on S. purpurea & U. gibba projects.
- Work on nanopore sequencing, genome assembly, functional annotation, construction of evolutionary history, and characterization of genome architecture for various mint and bear species.

Summer/Undergraduate Research Assistant 06/25 – 12/25
Principal Investigator(s): Michael Gore, Kelly Robbins
Plant Breeding and Genetics Section, Cornell University, Boyce Thompson Institute
Summer Program: CROPPS Programmable Plant Systems REU

- Construct and preprocess datasets from 2023 and 2024 field trials investigating the effects of slac1 mutations on maize stomatal conductance across genotypes and environmental conditions.
- Develop a longitudinal modeling pipeline combining Gaussian curve fitting, linear mixed-effects models, and random forest analysis in R to characterize genotype-dependent and environmental variation in maize stomatal conductance.
- Assist with field trials and imaging for Ruby tomato phenotyping.

Undergraduate Research Assistant 09/24 – 01/2026
Principal Investigator: Victor Albert
Department of Biological Sciences, University at Buffalo
Responsibilities:

- Assemble and scaffold the first draft genome of Sarracenia purpurea using long-read sequencing and Hi-C chromosomal interaction data, followed by assembly quality assessment (BUSCO, QUAST) and iterative polishing.
- Perform functional annotation and comparative genomic analyses to investigate the evolution of carnivory, including digestive enzyme gene families, genomic repeats, and other features of genome architecture associated with specialized digestive functions and morphology.

Publications

Albert, V. A., Kirshner, J., Page, C., **Pratt, N.**, Merkel, J., Richter, M., Freund, M., Fukushima, K., & Lindqvist, C. Genome evolution and convergent innovation in the carnivorous plant *Sarracenia purpurea*. *G3: Genes|Genomes|Genetics (in press)*. *bioRxiv preprint:* 10.64898/2025.12.26.696377

Honors

UB Excellence in Research, Scholarship and Creativity Award	2025
Barry M. Goldwater Scholarship	2025
Benjamin A. Gilman International Scholarship to Czech Republic	2024
UB Dean's List	2022 – 2026
McNair Scholar Program, University at Buffalo	2024 – 2026
Daniel Acker Scholars Program & Scholarship, University at Buffalo	2022 – 2026
NYS Triple C Award	2022

Presentations

Pratt, N. (2026, August). *Biophysics Constrained Models for Interpretable Representations of Regulatory Genomics*. Oral presentation as part of the National Institute for Theory and Mathematics in Biology MathBio Ideathon, Chicago, IL.

Pratt, N. (2026, May). *Identifying High Value Data to Improve Genome Foundation Models*. Poster presentation at the CSHL 90th Symposium on Quantitative Genetics: AI x Biology, Long Island, NY.

Pratt, N. (2025, October). *Longitudinal Modeling of Maize Gas Exchange for Optimizing Water Use Efficiency*. Oral presentation at the Gulf Coast Undergraduate Research Symposium, Houston, TX.

Pratt, N. (2025, August). *Genomic and Environmental Effects on Maize Gas Exchange*. Oral presentation at the Boyce Thompson Institute George and Helen Kohut Research Symposium, Ithaca, NY.

Pratt, N., Page, C. (2025, April). *Sarracenia purpurea: Genome Assembly and Analysis of the Carnivorous North American Purple Pitcher Plant*. Poster presentation at the Celebration of student excellence Research Showcase, Buffalo, NY.

Pratt, N. (2025, March). *Sarracenia purpurea: Hierarchical Convergence in Morphology and Genome Evolution of Carnivorous Pitcher Plants*. Poster presentation at the 26th Annual McNair Scholars Undergraduate Research Conference, College Park, MD.

Pratt, N. (2024, July). *Enhancing Nanopore Sequencing: Leveraging Machine Learning to Detect Oxidative DNA Lesions*. Poster presentation at the SUNY Chancellor's Biological Sciences Summer Undergraduate Research Excellence Program Symposium, Buffalo, NY.

Work Experience

Resident Assistant 08/25 – 06/26
University at Buffalo Campus Living, Buffalo, NY

- Support 40 residents by fostering a safe and academically focused living environment.
- Plan and lead community programming, mediate conflicts, and connect residents with university resources.

Project Manager/Estimator/Laborer 06/23 – 06/26
The MJA Company, Buffalo, NY

- Bid and estimate price construction projects, developed standardized proposal systems, coordinate crews and schedules, and ensure execution matches project plans and budgets.
- Work independently installing epoxy flooring and chemical floor penetrants, polishing and repair concrete flooring.

Teaching Experience

Teaching Assistant (Intermed. Comp. Stat Programming; Organismal Bio Lab) Spring 2027
University of California Merced, Merced, CA

Teaching Assistant & Substitute Lecturer (Intro to Data Science) Fall 2026
University of California Merced, Merced, CA

Academic Tutor (Stat 301/302, Bioinfo 419) Spring 2026
University at Buffalo, Buffalo, NY